

Productivity Prediction in The Garment Industry Using Supervised and Unsupervised Machine Learning Models with Explainable AI

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Introduction

Productivity prediction of garment employee through machine learning

The garment industry is one of the most labor-intensive industries in the world. About 75 million people are directly involved in the textile, clothing and footwear sector worldwide.(united, 2021). In Uganda, the market is projected to experience a volume growth of 3.7% in 2025.

Productivity prediction is crucial for

- resource allocation,
- cost optimization,
- production planning
- worker welfare.

Machine learning offers promising solutions for accurate productivity prediction.

Background and motivation

The garment industry faces significant challenges in aligning employee productivity with management-set targets, leading to production inefficiencies and losses. Variations in employee work pace and aptitude often hinder achieving desired productivity levels. Current practices, such as manual monitoring and analysis of production across diverse teams and departments, are labor-intensive and prone to inefficiencies.

This study explores machine learning techniques, including regression and classification models, to predict employee productivity. These automated approaches reduce the reliance on manual tracking, enabling more efficient analysis and decision-making to optimize productivity and achieve organizational goals.

Literature Review

- traditionally, an individual that supervises the employee or are their direct manager is the best person to conduct an employee performance evaluation.
- Muhammad et al. (2023) integrated physiological and behavioral features (e.g., stress) with machine learning models like Random Forest, XGBoost, and Logistic Regression, demonstrating significant impacts on office worker productivity.
- Time study, altering operations, and reducing machine waiting time improve operator efficiency and increase production.
- Zannatuf and Aftab (2023) used physical, social, and economic environmental factors combined with classification algorithms to predict employee performance and aid in decision-making

Gaps in Related work

- Many proposed models are black-box models, which lack explainability and interpretability. This makes it difficult for managers to trust and understand the decisions made by these models.
- There is limited research focusing on region-specific factors, particularly in areas like East Africa.
- Most existing models primarily explore supervised learning techniques for predicting employee productivity

Problem Statement

The garment industry, like any other labor-intensive industry, faces the common issue of employee productivity not meeting the actual productivity set by management, leading to significant losses. The use of machine learning to predict productivity of garment employees.

Significance of the Project

- **Accurate Productivity Forecasting using machine learning:** The project employs advanced regression techniques enhanced by machine learning, enabling precise predictions of employee productivity. This allows organizations to better anticipate performance outcomes and make informed decisions to optimize efficiency.
- **Pattern Discovery through Unsupervised Learning:** By integrating unsupervised learning, the project uncovers hidden patterns and clusters within the dataset. This exploratory analysis provides deeper insights into the factors influencing productivity, even in the absence of predefined labels.
- **Improved Interpretability of AI Models:** The use of explainable AI tools, specifically LIME, ensures transparency in how input features affect the model's predictions. This addresses the trust issues commonly associated with black-box algorithms, fostering confidence among stakeholders in the decision-making process.

ML Research Objectives

1. Identify key variables influencing employee performance in the garment industry.
2. Develop and adopt machine learning algorithms for predictive modeling of employee performance..
3. Evaluate and compare the predictive models to determine which is more accurate and applicable.

Research Questions

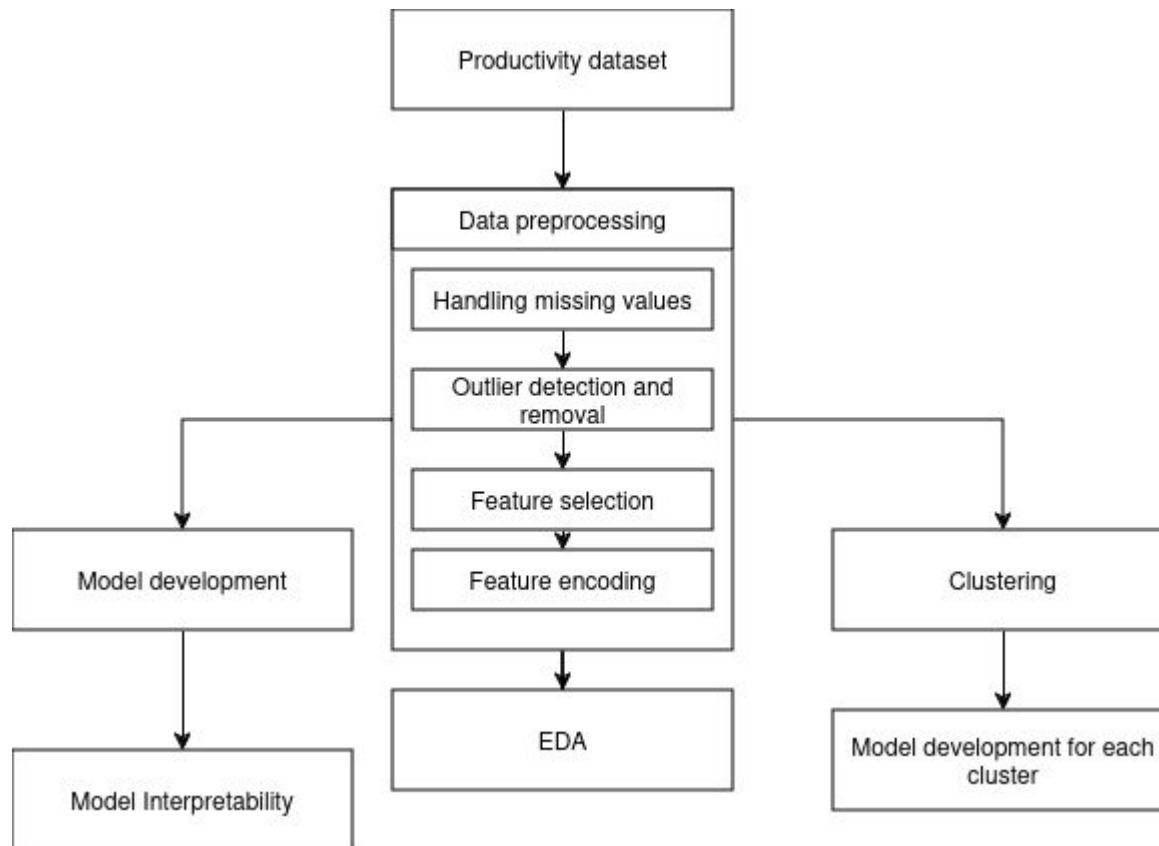
1. How can machine learning be adapted to create predictive models tailored to employee performance in the garment industry?
2. What factors and variables significantly influence the performance of employees in garment industry?
3. Which machine learning algorithms provide the most accurate and relevant predictions for employee performance?

Dataset Description

- Dataset: Garments worker productivity
- Source: UCI repository website
- Published: 2020
- Attributes: 15, including date, day, quarter, department, team_no, no_of_workers, no_of_style_change, targeted_productivity, SMV, wip, over_time, incentive, idle_time, idle_men, actual_productivity
- Actual Productivity Classes: Continuous
- Instances: 1197

Methodology

Flow diagram of your work



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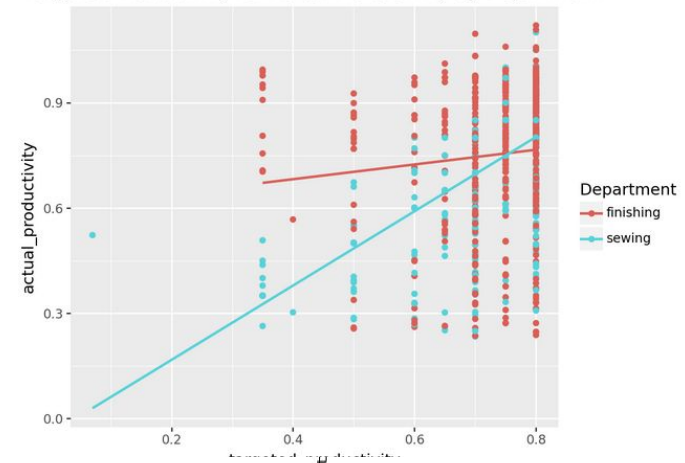
- EDA on the dataset involved the following
 - imputing missing values of 'wip' column
 - correcting miss-representations (spellings)
 - identifying outliers
 - Univariate analysis
 - Bivariate analysis
 - multivariate analysisusing plots to visualize the above analysis such as
 - bar plots, box plots, timeseries, histograms, heatmaps
- Model development using Multi-linear regression, random forest, lasso and XGBoost regression models
- Model evaluation to find the best performer
- Explainability of the models using lime
- clustering

Explanation

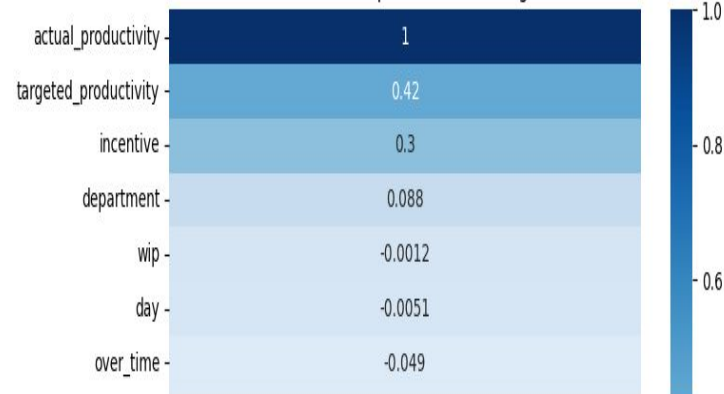
- The sewing department contributed more to the garment productivity compared to the finishing department. This is because the slope of the sewing department has a stronger linear relationship compared to the finishing department.

- Target productivity has a fairly positive correlation with the target (actual productivity) followed by incentive and department

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Targeted Productivity vs. Actual Productivity by Department

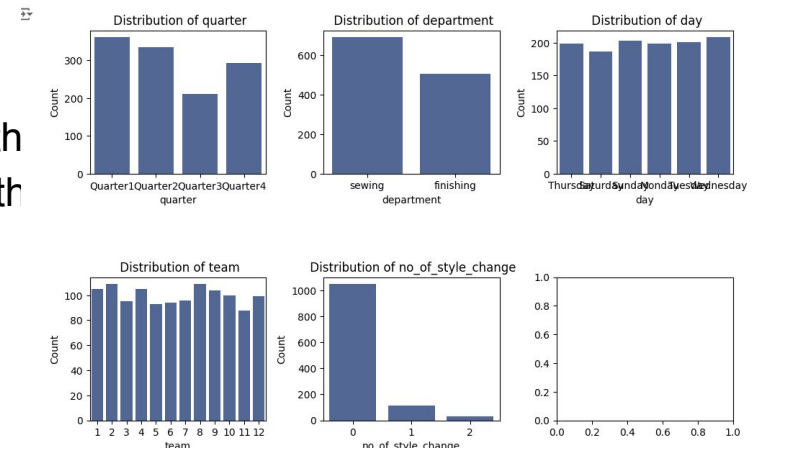


Correlations between predictors and target column

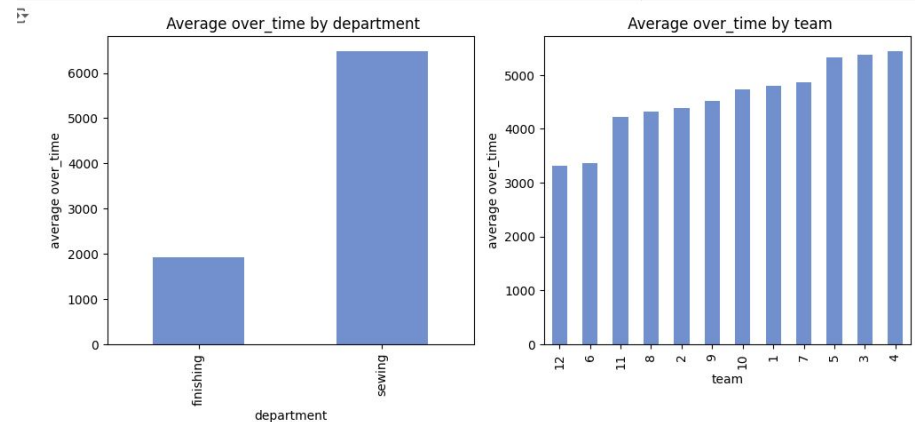


Explanation

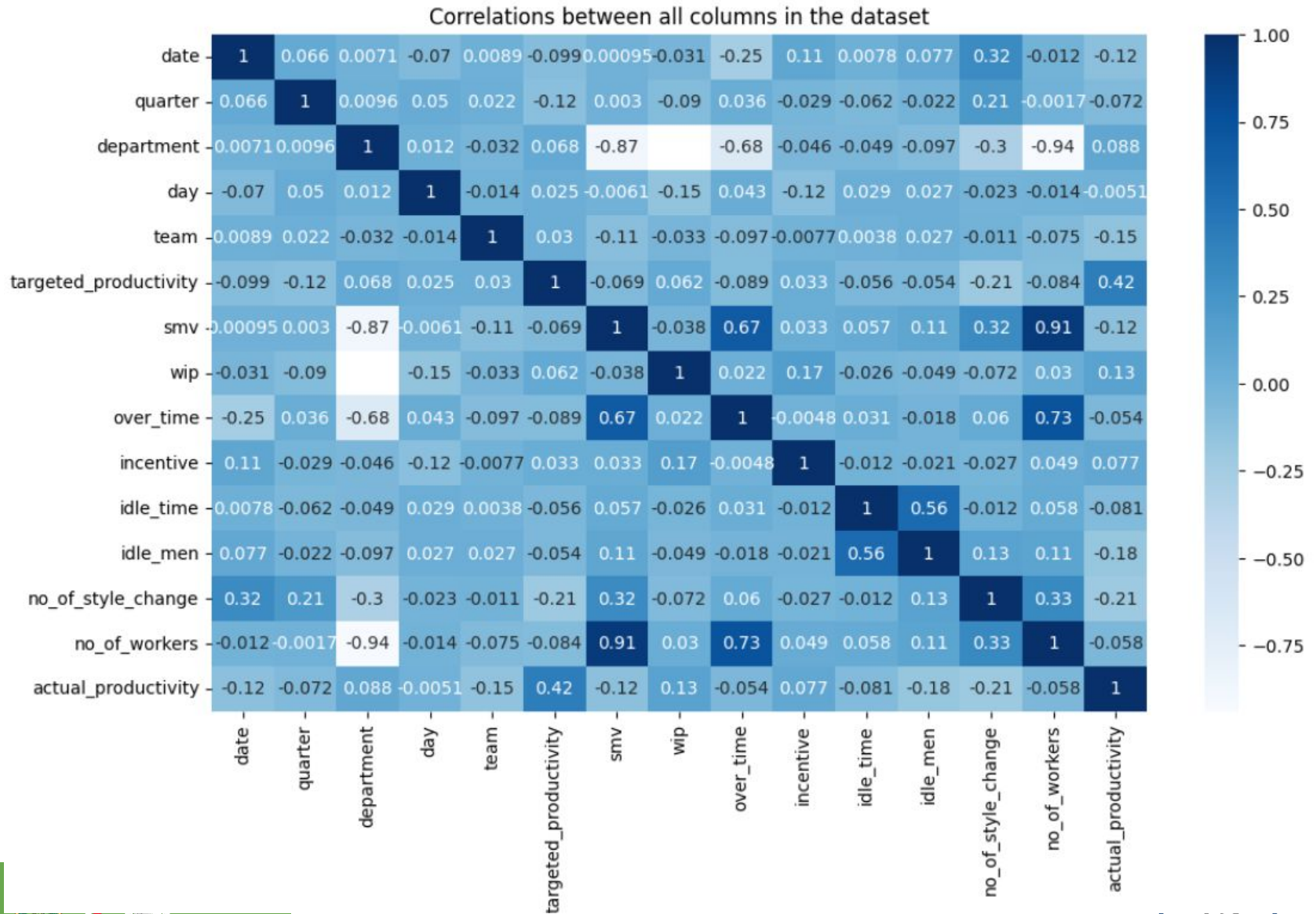
- wednesday has the most records followed by thursday
- quarter 1 had the most records compared to other quarters
- under teams, 2 and 8 had the most records.



- the sewing department recorded the highest overtime
- team 4 recorded the highest overtime



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Model Results

Data preparation for model development

- One hot encoding of the data
- splitting the data 0.2 test 0.8 train
- scaling the data to normalize the range of independent variables

Model Performance Metrics:

	Linear Regression	Lasso Regression	Random Forest	XGBoost
MAE	0.1074	0.1104	0.0887	0.0955
MSE	0.0228	0.0242	0.0213	0.0248
RMSE	0.1509	0.1554	0.1458	0.1576
R2 Score	0.2542	0.2084	0.3030	0.1859

Best Models for Each Metric:

MAE: Random Forest (0.0887)
MSE: Random Forest (0.0213)
RMSE: Random Forest (0.1458)
R2 Score: Random Forest (0.303)

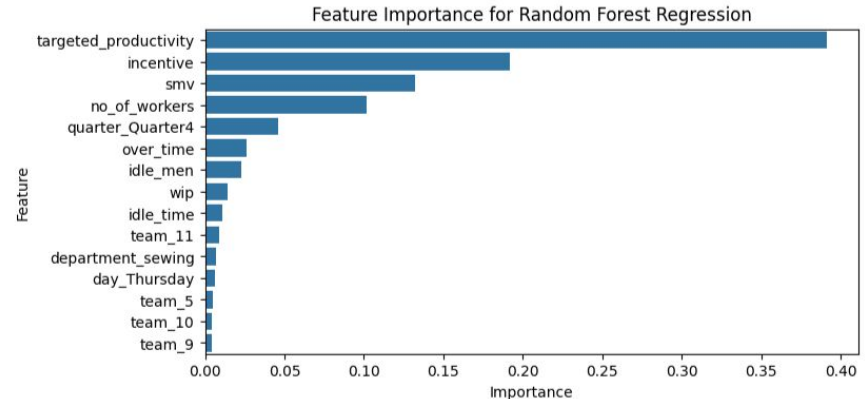
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- Top models; Random forest and XGBOOST
MAE value (0.088)
R2-score - 0.37
- After hyper parameter tuning the random forest gives an MAE value of 0.077

Random forest regression model is considered as the best model from this study since it has the lowest MAE value and the highest R2-score value.

Key Influencing Features in Productivity

- Most Important Features:
 1. Target Productivity – Highest
 2. Incentive
 3. Standard Minute Value (SMV)
 4. Number of Workers – Lowest

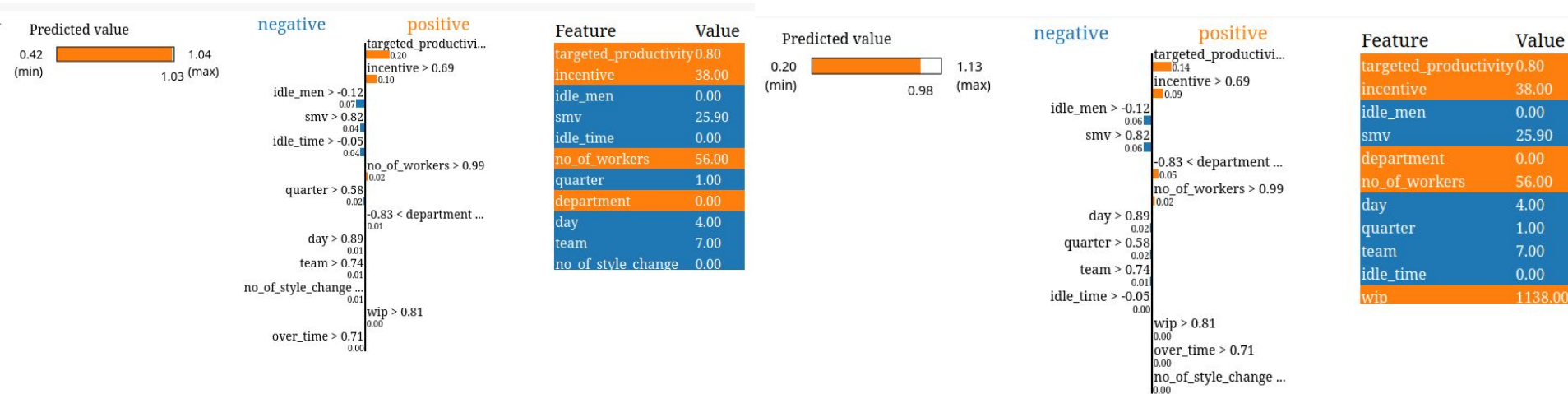


Explainability of the models using Lime

Model Comparison for First Instance

- **Linear Regression:**
 - Predicted productivity: -17.0 (actual lime value -8)
 - **Key Negative Influencers:** Target productivity, idle men, idle time, SMV
 - **Minor Positive Influencers:** Number of workers, department
- **Random Forest:**
 - Predicted productivity: 0.61 (actual lime value 0.57)
 - **Top Negative Influencers:** Idle men , SMV
 - **Positive Influencers:** No. of workers , department, incentive
- **XGBoost:**
 - Predicted productivity: 0.73 (actual lime value 0.73)
 - **Key Negative Influencers:** Idle men , SMV , target productivity, WIP
 - **Positive Influencers:** No. of workers, department

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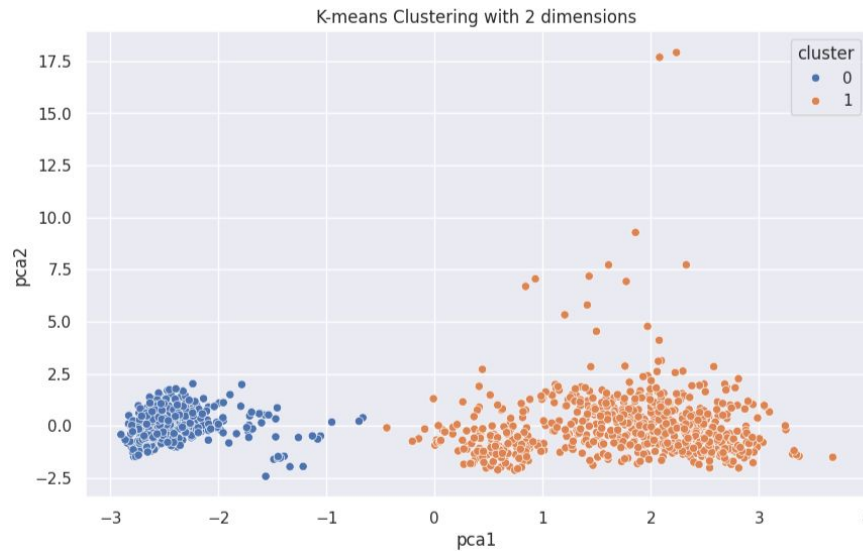
For 20th and 50th instances

Top models : random forest and xgboost provide better explanations with the lime productivity value close to the actual productivity value e.g (predic 0.50, actual 0.54)

- features influencing the productivity positively for the random forest include target productivity, incentive, no-of-workers

Clustering - unsupervised learning

Using Elbow method the number of clusters arrived at was 2.
below is a representation of the data points after clustering



Homogeneity: 1.0

Completeness: 1.0

Silhouette Score: 0.14

- The perfect homogeneity and completeness scores depict an excellent class-to-cluster mapping
 - The silhouette score indicates low inter-cluster separation

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R2-Score result for each cluster for each model

- cluster 1 shows the best r2 score with random forest and xgboost having the highest r2 score values.



	Linear Reg	Random Forest Regression	Lasso Reg	XGB regression
Cluster0	0.136808	0.232902	0.111500	0.357502
Cluster1	0.736697	0.811142	0.741899	0.808361

Conclusion and Future Work

The study successfully developed a machine learning framework for predicting employee productivity in the garment industry. Among the models tested, Random Forest emerged as the most reliable based on MAE evaluation and feature explainability, highlighting the significance of target productivity, incentives, workforce size, and departmental factors in driving employee performance.

However, the limited R^2 scores indicate a need for further refinement to improve the variance explained by the models.

Future work will focus on extending the application to East African datasets to localize the findings and deploying the best-performing model for practical use, enhancing productivity assessments in the region